



IDC216 : Statistics for Scientists (2025-26)

Exercise Set

1. The following data give the displacement (in cm) of a particle at equally spaced time intervals of 1 second:

$t(\text{s})$	0	1	3	4	5
$s(\text{cm})$	0	3	14	15	26

Use the binomial coefficient method of interpolation to estimate the displacement at $t = 2$ seconds. Compare your result when you interpolate using Newton's divided difference method.

2. The table below gives the velocity (m/s) of a car at different times:

$t(\text{s})$	0	2	4	6
$v(\text{m/s})$	0	12	35	80

Use Newton's divided difference interpolation to estimate the velocity at $t = 3$ s and $t = 5$ s. Then compute the approximate acceleration at $t = 4$ s using a suitable interpolating polynomial.

3. The boiling point ($^{\circ}\text{C}$) of a liquid is observed at different pressures (atm):

P	1.0	1.5	2.0	2.5
T	100.0	112.0	120.0	127.0

Fit a cubic polynomial $T = a + bP + cP^2 + dP^3$ to this data and predict the boiling point at $P = 1.8$ atm.

4. The population (in thousands) of a city was recorded as follows:

Year	2000	2005	2010	2015
Population	42	49	61	74

Construct an interpolating polynomial using Newton's divided difference method and estimate the population in the year 2012. Comment on the reliability of predicting for the year 2025 using the same polynomial.

5. The following table shows the stress-strain relationship for a mild steel specimen:

Strain	0.001	0.002	0.003	0.004
Stress (MPa)	100	200	300	380

- (a) Obtain the interpolating polynomial of degree 3.
(b) Estimate the stress at strain 0.0025.

6. The intensity of light (lux) at different distances (m) from a source is given as:

d	2	3	4	5
I	250	112	63	40

Does this data suggest that inverse-square law holds for intensity? If yes, compute the constant for this law.

7. The growth of bacterial colonies over time is recorded as:

$t(\text{hours})$	0	2	4	6
$N(\text{millions})$	1.0	2.2	5.0	9.2

Use a cubic interpolating polynomial to estimate the population at $t = 3$ h.

8. The following data represent measurements of current I (in amperes) for different voltages V (in volts) in a physics experiment:

$V(\text{V})$	1	2	3	4	5
$I(\text{A})$	0.52	1.05	1.47	2.00	2.42

- Fit a least squares regression line $I = a + bV$.
 - Compute the slope and interpret its physical meaning.
 - Compute R^2 to quantify the strength of the fit.
 - Identify which data point, if removed, changes R^2 the most and justify.
9. A biologist records the growth of plants (in cm) after applying varying amounts of fertilizer (g):

Fertilizer (g)	0	2	4	6	8	10
Height (cm)	10	14	18	22	24	27

- Fit a least squares line and compute regression coefficients.
 - Determine correlation coefficient r and coefficient of determination R^2 .
 - Interpret whether fertilizer quantity and plant height have a strong, moderate, or weak linear relationship.
10. In an environmental study, the concentration of particulate matter ($\text{PM}_{2.5}$, $\mu\text{g}/\text{m}^3$) and the number of asthma cases per 100,000 residents were recorded across 6 cities:

City	A	B	C	D	E	F
$\text{PM}_{2.5}$	30	45	60	70	80	85
Asthma Cases	122	148	190	214	228	242

- Perform linear regression of asthma cases on $\text{PM}_{2.5}$.
 - Compute the correlation coefficient and interpret its magnitude.
 - Discuss whether this relationship implies causation, and list two possible confounding variables.
 - Add a hypothetical “outlier” city with $(\text{PM}_{2.5}, \text{Cases}) = (85, 300)$ and recompute R^2 . Comment on the change.
11. The dataset below gives temperature (x , $^{\circ}\text{C}$) and enzyme activity (y , units/mL) in a biochemical reaction:

x	20	25	30	35	40	45
y	2.1	2.5	3.6	4.1	4.3	2.0

- Determine and interpret R^2 .
 - Discuss why linear regression would be inappropriate here.
12. A fair six-sided die is rolled. Let the random variable X represent the number shown on the die, and define a new variable $Y = X^2$.

- (a) Write down the probability distribution of X and compute $E[X]$ and $\text{Var}(X)$.
- (b) Compute $E[Y]$ and $\text{Var}(Y)$.
- (c) Verify numerically that $\text{Var}(aX + b) = a^2\text{Var}(X)$ for $a = 2$, $b = 1$.
- (d) Identify whether X and Y are independent random variables.
13. Let (X, Y) represent the outcome of tossing two fair coins, where $X = 1$ if the first coin shows head and 0 otherwise, and $Y = 1$ if the second coin shows head and 0 otherwise.
- (a) Write down the joint probability distribution table of (X, Y) .
- (b) Find the marginal distributions of X and Y .
- (c) Are X and Y independent? Verify mathematically.
- (d) Define a new random variable $Z = X + Y$ and find its distribution.
- (e) Compute $E[Z]$ and $\text{Var}(Z)$.
14. A bag contains 5 red, 3 blue, and 2 green balls. Two balls are drawn at random without replacement.
- (a) Define suitable random variables to represent the colors drawn.
- (b) Compute the probability that both balls are of the same color.
- (c) Compute the conditional probability that the second ball is red given that the first was blue.
- (d) Are the events “first ball is red” and “second ball is red” independent? Explain mathematically.
15. The heights (in cm) of adult males in a population are normally distributed with mean $\mu = 170$ and standard deviation $\sigma = 6$.
- (a) Find the probability that a randomly selected male is taller than 180 cm.
- (b) Find the 90th percentile of the height distribution.
- (c) Compute the probability that a male’s height lies between 160 cm and 175 cm.
- (d) If 9 individuals are chosen at random, what is the probability that their average height exceeds 172 cm? (Use the sampling distribution of the mean.)
16. A random variable X follows a discrete distribution given by:

x	0	1	2	3	4
$P(X = x)$	0.1	0.2	0.4	0.2	0.1

- (a) Compute $E[X]$, $\text{Var}(X)$, and $\text{SD}(X)$.
- (b) Find $P(X \geq 2)$.
- (c) Define a new variable $Y = 2X + 1$ and find its expectation and variance.
- (d) Suppose $Z = X + W$, where W is independent of X and uniformly distributed on $\{0, 1\}$. Compute $E[Z]$ and $\text{Var}(Z)$.
17. The weights (in grams) of packets of a chemical compound are normally distributed with mean 250 g and standard deviation 5 g.
- (a) If 100 packets are randomly sampled, find the approximate probability that the sample mean lies between 249 and 251 g.
- (b) Using the Central Limit Theorem, estimate the probability that the total weight of 100 packets exceeds 25,200 g.

(c) Discuss how the central limit theorem is crucial when the parent distribution is unknown or non-normal.

18. The following paired data show temperatures ($^{\circ}\text{C}$) and corresponding electricity consumptions (kWh) in a city over six days:

Temperature	20	22	25	27	30	32
Electricity Use	100	110	135	150	170	190

Compute the Pearson correlation coefficient between temperature and electricity use, interpret its sign and magnitude, and test for significance at the 1% level.

19. A psychologist records the stress level (on a 0–100 scale) and the number of hours of sleep per night for 10 subjects:

Sleep (h)	4	5	5	6	6	7	7	8	8	9
Stress	88	82	79	74	70	66	68	60	58	56

Compute both Pearson's r and Spearman's rank correlation coefficient. What do you conclude from these numbers.

20. The following table shows physics and mathematics test scores of 8 students:

Physics	55	60	65	70	75	80	85	90
Mathematics	52	62	66	68	72	77	83	88

Compute Pearson's correlation coefficient and the regression line of Mathematics on Physics. Predict the Mathematics score of a student who scored 68 in Physics.

21. In a survey of 200 students, responses to “Do you exercise regularly?” and “Do you smoke?” were recorded as follows:

	Smoker	Non-Smoker	Total
Exercise Regularly	25	85	110
Do Not Exercise	45	45	90
Total	70	130	200

Use a chi-square test of independence at 5% significance to determine whether smoking habit is independent of exercise habits.

22. The following contingency table summarizes survey data on gender and preference for online vs. in-person classes among 250 university students:

	Online	In-Person	Total
Male	60	80	140
Female	40	70	110
Total	100	150	250

Perform a chi-square test to examine whether class preference is independent of gender. Compute expected frequencies, chi-square statistic, and p -value.

23. A researcher studies relationship between blood type and susceptibility to a particular disease. In a sample of 400 individuals, the data are:

Blood Type	Disease	No disease	Total
<i>A</i>	35	65	100
<i>B</i>	45	55	100
<i>AB</i>	25	75	100
<i>O</i>	20	80	100
Total	125	275	400

Test whether disease incidence is independent of blood type at the 5% significance level. Comment on which blood group contributes most to the chi-square statistic.

24. A study records income level (Low, Medium, High) and preferred mode of transport (Public, Private) among 180 respondents:

	Public	Private	Total
Low	50	10	60
Medium	40	20	60
High	20	40	60
Total	110	70	180

Perform a chi-square test for independence and interpret whether transport choice depends on income level. Which cell contributes the largest component to the chi-square value?

25. In an ecological field study, the presence (Yes/No) of two plant species, A and B, was recorded across 100 sites:

	B Present	B Absent	Total
A Present	30	10	40
A Absent	20	40	60
Total	50	50	100

Test whether the occurrences of species A and B are independent at 5% level. Interpret the ecological meaning of your conclusion.

26. A social survey records data from 500 individuals on gender and their opinion on a new environmental policy (Support, Neutral, Oppose):

	Support	Neutral	Oppose	Total
Male	120	60	70	250
Female	130	80	40	250
Total	250	140	110	500

Test whether opinion on the policy is independent of gender using the chi-square test. Compute expected counts and interpret which category contributes most to the observed dependence.

27. A chemist claims that the mean concentration of a chemical solution is 0.5 mol/L. A random sample of 12 measurements yields a mean of 0.47 mol/L with a standard deviation of 0.06 mol/L. Formulate the null and alternative hypotheses to test the chemist's claim. Conduct a t-test at $\alpha = 0.05$ and interpret whether the claim is supported.
28. In a study of exam performance, the population mean score is believed to be 70. A new teaching method was tested on 10 students, yielding the following scores:

75, 72, 68, 74, 73, 78, 69, 71, 77, 70.

Test whether the new method significantly changes the mean score at the 5% level. Report the t-statistic, degrees of freedom, and p -value.

29. A researcher measures the mean blood pressure reduction for a drug in 8 patients as follows (mmHg): 10.2, 11.4, 9.5, 12.3, 10.8, 11.7, 9.9, 10.5. The claimed mean reduction is 10 mmHg. Perform a t-test at $\alpha = 0.01$. State whether to reject H_0 , and interpret the result in practical medical terms.

30. Two fertilizers, A and B, are tested on wheat yield (kg/acre). Data from 5 plots each are: A: [46, 49, 50, 47, 48], B: [52, 50, 55, 53, 51]. Test whether there is a significant difference between the two mean yields using a t-test at the 5% level. Compute 95% confidence interval for the difference in means.
31. In a physics lab, two instruments A and B measure the same physical quantity 8 times:
A: [10.1, 10.2, 10.0, 10.3, 10.1, 10.2, 9.9, 10.1], B: [10.2, 10.4, 10.1, 10.5, 10.3, 10.4, 10.2, 10.3]
Perform a paired t-test to determine if the two instruments differ significantly in their readings. Report t , degrees of freedom, and the p -value.
32. A psychologist compares stress scores of two groups of residents of a colony: one practicing a mindfulness program and one control group. Mindfulness: [20, 18, 22, 17, 19, 21, 23, 20] Control: [25, 28, 30, 24, 27, 29, 26, 31]. Test whether the mindfulness group has significantly lower stress at $\alpha = 0.05$. Do you think further information in the question would have helped you for having more confidence on your findings? What are these?
33. Revise the following:
- (a) null hypothesis,
 - (b) alternative hypothesis,
 - (c) p -value,
 - (d) significance level,
 - (e) the difference between Type I and Type II errors with an example.

Then, consider the following claim:

“There is no association between caffeine intake and alertness.”

Identify H_0 , H_a , and describe what rejecting or failing to reject H_0 means in this context.

34. Suppose an environmental scientist tests whether a water sample exceeds a safe limit of lead concentration ($3.5 \mu\text{g/L}$). Define the two types of errors in this setting and discuss which is more serious from a public health perspective.
35. Three different diets (A, B, C) are compared for average weight loss (kg) over 6 weeks. Data: A: [4.0, 3.8, 4.2, 3.9, 4.1] B: [3.0, 2.8, 3.2, 3.1, 3.0] C: [2.5, 2.6, 2.8, 2.4, 2.7]. Perform a one-way ANOVA to test whether the mean weight losses differ among diets. Compute the between-group and within-group variances, F -statistic, and interpret.
36. In an agricultural experiment, four fertilizers were tested on crop yield (kg/acre). Data are summarized as:

Fertilizer	n	Mean Yield	s^2
A	5	45.0	4.0
B	5	48.0	6.0
C	5	46.0	5.0
D	5	49.5	4.5

Perform an ANOVA using the summarized data. Compute the total, between-group, and within-group sums of squares. Test at $\alpha = 0.05$ whether the fertilizers differ in effect.

37. A research team tests whether the average reaction time (in seconds) differs across three age groups (Young, Middle-aged, and Elderly). The summary is given below:

Group	n	$\bar{x}$ (seconds)	s (seconds)
Young	20	0.45	0.05
Middle-aged	20	0.48	0.04
Elderly	20	0.53	0.05

Using a one-way ANOVA, compute the test statistic F and conclude whether age affects reaction time at the 1% significance level.

38. Case Study 1: The Miracle Fertilizer

A company launches a new “miracle fertilizer” claiming to increase crop yield by 25%. To test it, a local farm applies the fertilizer to 10 plots, each next to a control plot of similar soil and crop type. The results show that treated plots yield, on average, 10% more. However, later it’s revealed that all the treated plots were at a location which receives more sunlight, while control plots were in shaded areas.

Questions:

- Identify the treatment and control groups. Was proper randomization applied?
- What type of bias does this setup illustrate?
- Suggest a better experimental design to test the fertilizer’s effectiveness.
- If the company used these results in advertisements, discuss the ethical issues involved.

39. Case Study 2: A Sleep Study

A sleep researcher claims that playing soft music before bedtime improves sleep quality. They recruit 20 volunteers, ask them to listen to music for a week, and measure their average sleep duration. The mean sleep time increases from 6.2 hours (before) to 6.6 hours (after). The researcher publishes the result as statistically significant ($p < 0.05$). Months later, it is found that the researcher had actually measured *six different sleep variables*: total time, REM duration, deep sleep proportion, number of awakenings, latency, and efficiency, *but* only reported the one variable that showed significance.

Questions:

- Explain which bias or unethical practice this case exemplifies.
- How could replication and pre-registration have prevented the issue?
- Reframe how the study could be ethically reported, even if the result was not significant.

40. Case Study 3: The Mouse Memory Drug

A neuroscience team tests a new compound claimed to improve memory. Thirty mice are divided into two groups: 20 receive the compound, 10 get a placebo. After three weeks, treated mice complete a maze faster on average (mean time 18.5 s vs. 21.3 s). The researchers conclude that the drug enhances memory. Later analysis reveals that 5 slow-performing treated mice were removed from the data “because they seemed distracted”, and maze lighting was brighter in the treatment room.

Questions:

- Identify the major flaws in experimental design.
- What type of bias was introduced by selectively removing data points?

- (c) Suggest how replication, blinding, and balanced randomization could improve the validity.
- (d) Discuss why selective data omission violates data ethics, even if the drug's effect might be real.

41. Case Study 4: The Climate Prediction Shortcut

A data scientist models the relationship between CO₂ levels and global temperature increase using data between 1980-2020. They find a strong linear trend and extend it to forecast temperatures in the year 2200, predicting a catastrophic +12°C rise. A reviewer points out that the model ignores oceanic feedback cycles, energy balance limits, and saturation effects, making such extrapolation unreliable. The researcher defends the model, saying: "It's just an extension of the same straight line."

Questions:

- (a) Identify the misuse of extrapolation in this case.
- (b) Discuss how model validation and domain knowledge could have prevented the error.
- (c) Explain why extrapolation beyond the data range is not just a technical issue but an ethical one.
- (d) If this result were used to influence public policy, what safeguards should the scientific community enforce?